

Occasionally, the bowels of the internet even produce a scientific discovery that creeps into the world outside the purview of scientific journals or the “establishment.” In 2011, an anonymous poster on the online image board 4chan proposed and provided code for a sorting algorithm that had never been published before and has since come to be called *sleep sort*.

Sleep sort wasn’t designed to resemble any real-world situation, like inserting files into a filing cabinet. If we’re seeking an analogy, we might consider the task of allocating lifeboat spots on the *Titanic* as it began to sink. We might want to allow children and younger people the first chance to get on the lifeboats, and then allow older people to try to get one of the remaining spots. If we make an announcement like “younger people get on the boats before older people,” we’d face chaos as everyone would have to compare their ages—they would face a difficult sorting problem amidst the chaos of the sinking ship.

A sleep-sort approach to the *Titanic* lifeboats would be the following. We would announce, “Everyone please stand still and count to your age: 1, 2, 3, As soon as you have counted up to your current age, step forward to get on a lifeboat.” We can imagine that 8-year-olds would finish their counting about one second before the 9-year-olds, and so would have a one-second head start and be able to get a spot on the boats before those who were 9. The 8- and 9-year-olds would similarly be able to get on the boats before the 10-year-olds, and so on. Without doing any comparisons at all, we’d rely on individuals’ ability to pause for a length of time proportional to the metric we want to sort on and then insert themselves, and the sorting would happen effortlessly after that—with no direct inter-person comparisons.

This *Titanic* lifeboat process shows the idea of sleep sort: allow each element to insert itself directly, but only after a pause in proportion to the metric it’s being sorted on. From a programming perspective, these pauses are called *sleeps* and can be implemented in most languages.

In Python, we can implement sleep sort as follows. We will import the `threading` module, which will enable us to create different computer processes for each element of our list to sleep and then insert itself. We’ll also import the `time.sleep` module, which will enable us to put our different “threads” to sleep for the appropriate length of time.